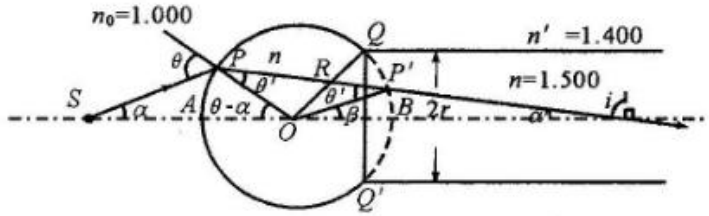


$$\sin 30° = \frac{R}{\overline{SO}}$$

$$\overline{SA} = R$$

$$x \leqslant R \qquad (1)$$

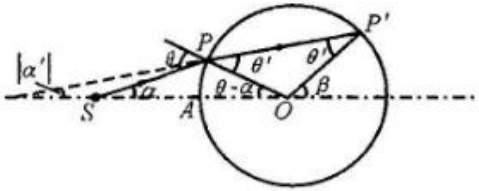
$$\frac{2R}{\sin\theta} = \frac{R}{\sin\alpha} \qquad (2)$$



$$\sin\theta = 1.5\sin\theta' \qquad (3)$$

$$\beta = 2\theta' - \theta + \alpha \qquad (4)$$

$$\alpha' = \theta - \alpha - \theta' \qquad (5)$$



$$|\beta| \leqslant \angle QOB \qquad (6)$$

$$\angle QOB = \arcsin\frac{r}{R}$$